

20250922EB_MN (seahorse, vLH103, vLH78)

Project: Yi-Shuan Tseng Holt Lab
Author: Yi Shuan Tseng
Status: In progress
Entry Created On: 13 Oct 2025 19:59:56 UTC
Entry Last Modified: 16 Oct 2025 02:12:15 UTC
Export Generated On: 01 Sep 2026 18:34:00 UTC

- 2025年9月22日 星期一
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2025年9月24日 星期三
- T2 -> T3
- 2025年9月26日 星期五
- T3 -> T4 (~14 mL for over-weekend)
- 2025年9月29日 星期一
- T4 -> T5
- 2025年10月1日 星期三
- T5 -> T6
- 2025年10月3日 星期五
- T6 -> T6 (~14mL for over-weekend)
- 2025年10月6日 星期一
- T6 -> T7
Dissociation
 1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
 2. Remove the PLO and wash the plate with PBS once
 3. Coat the plate with laminine for 1hr.
 4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
 5. Add 2.5mL accutase and tap the cells
 6. Put the tube in RT (try to incubate in RT since accutase has higher activity at lower temp). Flick the tube every 5min until 40-50min, but the EBs looked very intact. I pipetted it up and down for 30 times, but the EBs were still there. I transfer the supernatant to process the following steps and add 1mL new accutase to remaining EBs.

7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently.
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 200rcf, 3.5min. Remove the supernatent. Add 1mL of T7 and pipette it gently and thoroughly.
11. Add 4mL more T7 to resuspend the cells and count it.
12. The yield was too low, so I processed batch 2 after 30min incubation of new 1mL accutase -> 1ml HI-FBS and CTWM -> filter -> centrifuge
13. Plate the cells with T7.

cell info				
	A	B	C	D
1	cell	passage	well#	note
2	WT11	P4	5	This batch of iPSCs haven't grown on corning DMEM matrigel, but I still have issue with dissociation.

Table1		
	A	B
1	run1	volume (mL)
2	trypsin	3
3	HI-FBS/CTWM	6
4	run2	*take some undissolved EB
5	trypsin	2
6	HI-FBS/CTWM	4

Table3									
	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	4.26	99	7	29.82	A	96	60	36	507
3					B	24	250	16	939
4					C,D	6	1972	12	the rest

PlateA

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B		60K			90K			125K				
C												
D												
E												
F												
G												
H												

PlateB

	1	2	3	4	5	6
A						
B	103	103	78			
C	103	103	78			
D	103	78	78			

2025年10月7日 星期二

PlateB: Change T7 -> T7 if added virus

2025年10月8日 星期三

PlateA: T7-> T7 if not added virus

2025年10月9日 星期四

T7 -> T8

2025年10月11日 星期六

1/2 T8 -> 1/2 T8

2025年10月13日 星期一

1/2 T8 -> 1/2 T8

2025年10月14日 星期二

1/2 T8 -> 1/2 T8

Treat cells for 24h hour group (For plate B, it would be less than 24h. Would be round 20h). All the treatments are Neurobasal-A media with 1mM pyruvate, glutaMAX and full supplements.

Treat Plate A at 6:37pm (row B: 2.5mM glucose; rowC: 0.15mM glucose)

Treat Plate B at 6:50pm (B1, B2, B3: 2.5mM glucose; C2, C3: 0.15mM glucose)

2025年10月15日 星期三

Seahorse Plate A

condition: 1.5uM oligomycin, 1uM FCCP, 0.5uM Rot/AA

1. Rehydrate the cartridge the day before.

2. Turn on the seahorse machine.

3. Prepare drugs from stock as the drug table below with two glucose concentrations.

4. every well wash with 90-100uL media with corresponding glucose concentration

5. add 175uL of media to the cells and add 2.5mM glucose media to all the blank wells.

6. incubate the plate for 1 hour to deplete CO2.

7. add drug to the cartridge as below. Add H2O for all the remaining wells.

drug in cartridge layout			^
	A	B	
1	A (oligo)	B (FCCP)	
2	C (Rot/AA)	D (H2O)	

drug calculation per glucose concentration																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1		stock (uM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution	working (uM)	loading volume	well volume before loaded	loading dilution	loading conc.(uM)	dilution factor from 1st dil.	well# for each condition	total loading volume	volume of 1st dil.	volume of media
2	oligo	10000	10	20	2	18	1000	1.5	25	175	8	12	83.33333333333	18	500	6.0	494.0
3	FCCP	10000	20	20	1	19	500	1	25	200	9	9	55.55555555556	18	500	9.0	491.0
4	Rot/AA	10000	10	10	1	9	1000	0.5	25	225	10	5	200	18	500	2.5	495.0
5	2DG	2000000					2000000	50000	25	250	11	550000	3.6363636364	18	500	137.5	225.0
6	medium								275					18	5500		5500
7						46								10500			6980
8																	7026

drug calculation per glucose concentration1								^
	A	B	C	D	E	F	G	
1		stock (mM)	working (mM)	loading dilution	total loading volume	volume of media		
2	glucose	1000	2.5	400	18000	45.0	17595	
3		1000	0.15	6666.6 666666 667	8000	1.2	7838.8	
4	glutamine	200	4	50	18000	360.0		
5		200	4	50	8000	160.0		
6								
7								

Well2													^
	1	2	3	4	5	6	7	8	9	10	11	12	
A													
B		2.5mM glucose 20h											
C		0.15mM glucose 20h											
D		2.5mM glucose 1h											
E		0.15mM glucose 1h											
F													
G													
H													

Cell counting (normalization) using total protein concentration

Imaging nucGEM (PlateB)
100x new spinning disk confocal: 0.11um/px

Well1							^
	1	2	3	4	5	6	
A							
B	3	3	3				
C	1	4	4				
D	2	1	2				

Table4									/
	A	B	C	D	E	F	G	H	
1			vLH78			vLH103			
2	condition#	condition	treat time	SirDNA only	image time	treat time		image time	
3	1	2.5mM glucose 1h	20:50		21:58	19:30		20:28	
4	2	0.15mM glucose 1h	21:10		22:11	19:55		20:55	
5	3	2.5mM glucose 24h	18:50 10/14	18:45	21:30	18:45 10/14	18:45	19:49	
6						18:45 10/14	18:45	20:00	
7	4	0.15mM glucose 24h	18:50 10/14	18:45	21:43	18:45 10/14	18:45	20:15	

2.5mM: 5 wells -> 400*5 = 2000 2000/5000=0.4
0.15mM: 4 wells -> 400*4 = 1600 1600/5000= 0.32

dilute 1:20 of SirDNA -> 0.72 in 13.7uL 0.15mM media --> 1.6 uL/well

103_O.C. , 100X, 1xzoom, 600x900, 12-bit						^
	A	B	C	D	E	
1	channel	bit	power (%)	exposure time	note	
2	488	12	100	10ms		
3	640	12	15	100 ms	snapshot	
4	488	12	20	100 ms		

Table2

	A	B	C	D
1		treatment	treat time	type
2	vLH103_MN_100x_1xzoom_0001	2.5mM gluc	24h	movie
3	vLH103_MN_100x_1xzoom_0002	2.5mM gluc	24h	snapshot
4	vLH103_MN_100x_1xzoom_0003	2.5mM gluc	24h	movie
5	vLH103_MN_100x_1xzoom_0004	2.5mM gluc	24h	snapshot
6	vLH103_MN_100x_1xzoom_0005	0.15mM gluc	24h	movie
7	vLH103_MN_100x_1xzoom_0006	0.15mM gluc	24h	snapshot
8	vLH103_MN_100x_1xzoom_0007	2.5mM gluc	1h	movie
9	vLH103_MN_100x_1xzoom_0008	2.5mM gluc	1h	snapshot
10	vLH103_MN_100x_1xzoom_0009	0.15mM gluc	1h	movie
11	vLH103_MN_100x_1xzoom_0010	0.15mM gluc	1h	snapshot
12	vLH78_MN_100x_1xzoom_0001	2.5mM gluc	24h	movie
13	vLH78_MN_100x_1xzoom_0002	2.5mM gluc	24h	snapshot
14	vLH78_MN_100x_1xzoom_0003	0.15mM gluc	24h	movie
15	vLH78_MN_100x_1xzoom_0004	0.15mM gluc	24h	snapshot
16	vLH78_MN_100x_1xzoom_0005	2.5mM gluc	1h	movie
17	vLH78_MN_100x_1xzoom_0006	2.5mM gluc	1h	snapshot
18	vLH78_MN_100x_1xzoom_0007	0.15mM gluc	1h	movie
19	vLH78_MN_100x_1xzoom_0008	0.15mM gluc	1h	snapshot

